

ON THE UNIQUENESS OF THE EVEN PRIME

ABSTRACT. We prove that 2 is the unique even prime natural number, in a development that begins from the Dedekind–Peano axioms and constructs the arithmetic and order needed along the way. Recursion is established before the order relation is defined. Addition and multiplication are constructed, and every identity later required—including additive cancellation, the impossibility of a nonzero summand yielding zero, and the fact that the only unit is 1—is proved explicitly. After defining primes, composites, and even numbers, we show that nothing strictly less than 2 is prime and that every even natural number strictly greater than 2 is composite. The latter is proved in two arithmetically distinct ways: by the divisor clause in the definition of primality, and by exhibiting a proper factorization. A companion LEAN 4 development checks the uniqueness layer on the kernel type `Nat`; we record the precise trust boundary between that artifact and the Peano narrative.

1. INTRODUCTION

Theorem 1.1 (Main theorem). *Every even prime natural number equals 2. Moreover 2 itself is an even prime, so 2 is the unique even prime.*

The first sentence is a uniqueness-only claim; the second adds the witness that 2 is an even prime. Both are proved below. Neither is deep, but a fully explicit account must specify what a natural number is, construct arithmetic, and justify every intermediate step. That is the purpose of this note.

We follow the classical programme of Landau’s *Foundations of Analysis* [1], with three design choices of our own. First, recursion is proved before order is introduced, so that the construction of addition does not depend on a relation that itself depends on addition. Second, we carefully separate two genuinely different proofs that every even $n > 2$ is composite from mere presentational variants of a single inference. Third, a companion formalization in LEAN 4 checks the uniqueness layer on `Nat`; [Section 8](#) states exactly what is and is not machine-checked.

The development occupies [Sections 2](#) to [5](#). Primes and the uniqueness argument occupy [Sections 6](#) and [7](#).

2. AMBIENT FRAMEWORK AND THE PEANO AXIOMS

Convention. We work in informal classical set theory. We allow formation of subsets of $\mathbb{N}$, functions $\mathbb{N} \rightarrow \mathbb{N}$, and subsets of $\mathbb{N} \times \mathbb{N}$. Induction is stated for arbitrary subsets (Dedekind’s form). Classical logic is used freely.

Definition 2.1 (Natural numbers). We postulate a set $\mathbb{N}$, a distinguished element $0 \in \mathbb{N}$, and a function $S: \mathbb{N} \rightarrow \mathbb{N}$, subject to the axioms below.

Axiom 2.2. For every $n \in \mathbb{N}$, $S(n) \neq 0$.

Axiom 2.3. For all $m, n \in \mathbb{N}$, if $S(m) = S(n)$, then $m = n$.

Axiom 2.4 (Induction). Let $A \subseteq \mathbb{N}$. If $0 \in A$ and $n \in A \Rightarrow S(n) \in A$ for every n , then $A = \mathbb{N}$.

Notation 2.5. Set $1 \stackrel{\text{def}}{=} S(0)$, $2 \stackrel{\text{def}}{=} S(1)$, $3 \stackrel{\text{def}}{=} S(2)$, and $4 \stackrel{\text{def}}{=} S(3)$.

Date: Version 3, July 30, 2026.

2020 Mathematics Subject Classification. Primary 11A41, 03B30; Secondary 00A35.

Key words and phrases. Prime number, even integer, Peano axioms, recursion theorem, divisibility, elementary number theory.

Lemma 2.6. *One has $0 \neq 1$ and $1 \neq 2$.*

Proof. If $0 = 1 = S(0)$, contradict [Axiom 2.2](#). If $1 = 2$, then $S(0) = S(1)$, so $0 = 1$ by [Axiom 2.3](#). $\square$

Lemma 2.7 (Predecessor). *For every $n \in \mathbb{N}$, either $n = 0$ or there is a unique m with $n = S(m)$.*

Proof. Let A be the set of n satisfying the claim. Then $0 \in A$, and if $n \in A$ then $S(n) \in A$ by taking $m = n$. By [Axiom 2.4](#), $A = \mathbb{N}$. Uniqueness is [Axiom 2.3](#). $\square$

3. RECURSION, ADDITION, AND MULTIPLICATION

Lemma 3.1 (Recursion). *Let $a \in \mathbb{N}$ and $f: \mathbb{N} \rightarrow \mathbb{N}$. There exists a unique $u: \mathbb{N} \rightarrow \mathbb{N}$ such that*

$$u(0) = a \quad \text{and} \quad u(S(n)) = f(u(n)) \quad \text{for all } n \in \mathbb{N}. \quad (3.1)$$

Proof. Uniqueness of u . If u and v both satisfy (3.1), the set $\{n : u(n) = v(n)\}$ contains 0 and is closed under S , so equals $\mathbb{N}$ by [Axiom 2.4](#).

Existence. Call $s \subseteq \mathbb{N} \times \mathbb{N}$ a *recursive sequence* for (a, f) if $(0, a) \in s$ and $(n, y) \in s \Rightarrow (S(n), f(y)) \in s$. The collection $\mathcal{S}$ of all recursive sequences is nonempty, since $\mathbb{N} \times \mathbb{N} \in \mathcal{S}$. Set $\Gamma \stackrel{\text{def}}{=} \bigcap_{s \in \mathcal{S}} s$. Then Γ is itself a recursive sequence, and it is the least one.

We claim that for every n there is a unique y with $(n, y) \in \Gamma$.

Existence of y . Let $A = \{n : \exists y (n, y) \in \Gamma\}$. Then $0 \in A$. If $n \in A$ with $(n, y) \in \Gamma$, then $(S(n), f(y)) \in \Gamma$, so $S(n) \in A$. Thus $A = \mathbb{N}$.

Uniqueness of y , and the predecessor property. We first show: if $(S(n), z) \in \Gamma$, then $z = f(y)$ where y is the unique value with $(n, y) \in \Gamma$ (assuming uniqueness at n for the moment). Indeed, if $z \neq f(y)$, then $\Gamma \setminus \{(S(n), z)\}$ still contains $(0, a)$ and is closed under the recursive step: the only way to produce a pair at $S(n)$ from a pair at n is to form $(S(n), f(y))$, which is retained. By injectivity of S ([2.3](#)), no other predecessor index equals n . Thus $\Gamma \setminus \{(S(n), z)\}$ is a recursive sequence, contradicting minimality of Γ .

Now let A be the set of n for which at most one y has $(n, y) \in \Gamma$. Then $0 \in A$: if $(0, y) \in \Gamma$ and $y \neq a$, remove $(0, y)$ as above. If $n \in A$ and $(S(n), z_1), (S(n), z_2) \in \Gamma$, the predecessor property forces $z_1 = z_2 = f(y)$ for the unique y at n . So $S(n) \in A$, and $A = \mathbb{N}$.

Define $u(n)$ to be the unique y with $(n, y) \in \Gamma$. Since Γ is a recursive sequence, $u(0) = a$ and $u(S(n)) = f(u(n))$. $\square$

Definition 3.2 (Addition and multiplication). For each m , apply [Lemma 3.1](#) with $a = m$ and $f = S$ to obtain u_m , and set $m + n \stackrel{\text{def}}{=} u_m(n)$. Thus

$$m + 0 = m, \quad m + S(n) = S(m + n). \quad (3.2)$$

For each m , apply [Lemma 3.1](#) with $a = 0$ and $f(x) = x + m$ to obtain v_m , and set $m \cdot n \stackrel{\text{def}}{=} v_m(n)$. Thus

$$m \cdot 0 = 0, \quad m \cdot S(n) = (m \cdot n) + m. \quad (3.3)$$

Lemma 3.3 (Zero on the left; successor as $+1$). *For every n , one has $0 + n = n$ and $n + 1 = S(n)$.*

Proof. For $0 + n = n$: induct on n . The case $n = 0$ is (3.2). Assuming $0 + n = n$, we have $0 + S(n) = S(0 + n) = S(n)$. For $n + 1$: $n + 1 = n + S(0) = S(n + 0) = S(n)$. $\square$

Lemma 3.4 (Associativity of addition). *For all a, b, c , one has $(a + b) + c = a + (b + c)$.*

Proof. Fix a, b and induct on c . For $c = 0$ both sides equal $a + b$. Assuming the claim for c ,

$$(a + b) + S(c) = S((a + b) + c) = S(a + (b + c)) = a + S(b + c) = a + (b + S(c)).$$

$\square$

Lemma 3.5 (Commutativity of addition). *For all m, n , one has $m + n = n + m$.*

Proof. First, $m + 1 = 1 + m$ for all m : induct on m . The case $m = 0$ is [Lemma 3.3](#). Assuming $m + 1 = 1 + m$,

$$S(m) + 1 = S(S(m)) = S(m + 1) = S(1 + m) = 1 + S(m).$$

Now fix m and induct on n . The case $n = 0$ is [Lemma 3.3](#) with (3.2). Assuming $m + n = n + m$,

$$\begin{aligned} m + S(n) &= S(m + n) = S(n + m) = n + S(m) = n + (m + 1) \\ &= n + (1 + m) = (n + 1) + m = S(n) + m, \end{aligned}$$

using [Lemma 3.4](#) and the claim of the previous paragraph. $\square$

Lemma 3.6 (Successor on the left). *For all m, n , one has $S(m) + n = S(m + n)$.*

Proof. By [Lemmas 3.3](#) and [3.5](#) and (3.2), $S(m) + n = (m + 1) + n = m + (1 + n) = m + S(n) = S(m + n)$. $\square$

Lemma 3.7 (Multiplication by one). *For every n , one has $n \cdot 1 = n$ and $1 \cdot n = n$.*

Proof. First, $n \cdot 1 = n \cdot S(0) = (n \cdot 0) + n = 0 + n = n$ by (3.3) and [Lemma 3.3](#). For $1 \cdot n$, induct on n : the case $n = 0$ is (3.3); assuming $1 \cdot n = n$, $1 \cdot S(n) = (1 \cdot n) + 1 = n + 1 = S(n)$. $\square$

Lemma 3.8 (Additive cancellation). *If $x + c = y + c$, then $x = y$. Consequently, if $a + k = a$, then $k = 0$.*

Proof. Induct on c . The case $c = 0$ is immediate. If $x + S(c) = y + S(c)$, then $S(x + c) = S(y + c)$, so $x + c = y + c$ by [Axiom 2.3](#), hence $x = y$ by induction. For the consequence: $a + k = a + 0$ implies $k = 0$. $\square$

Lemma 3.9 (No nonzero sum is zero). *If $a \neq 0$, then $a + b \neq 0$ for every b . In particular, if $a + b = 0$, then $a = b = 0$.*

Proof. Write $a = S(a')$ by [Lemma 2.7](#). Then $a + b = S(a') + b = S(a' + b) \neq 0$ by [Lemma 3.6](#) and [Axiom 2.2](#). The second claim follows. $\square$

Lemma 3.10 (No zero-divisors). *If $a \neq 0$ and $a \cdot b = 0$, then $b = 0$.*

Proof. Induct on b . If $b = 0$ we are done. If $b = S(b')$, then $a \cdot b = (a \cdot b') + a$. Write $a = S(a')$; then the sum equals $S((a \cdot b') + a') \neq 0$ by [Lemma 3.9](#), a contradiction. $\square$

Lemma 3.11 (Successor on the left in a product). *For all m, n , one has $S(m) \cdot n = m \cdot n + n$.*

Proof. Fix m and induct on n . For $n = 0$ both sides are 0. Assuming the claim for n ,

$$\begin{aligned} S(m) \cdot S(n) &= S(m) \cdot n + S(m) = (m \cdot n + n) + S(m) = m \cdot n + (n + S(m)) = m \cdot n + S(n + m) \\ &= m \cdot n + S(m + n) = m \cdot n + (m + S(n)) = (m \cdot n + m) + S(n) = m \cdot S(n) + S(n), \end{aligned}$$

using [Lemmas 3.3](#) to [3.5](#) and (3.3). $\square$

Lemma 3.12 (Units). *If $a \cdot b = 1$, then $a = b = 1$.*

Proof. Neither factor is 0, else the product is $0 \neq 1$. Write $b = S(b')$. Then $a \cdot b = (a \cdot b') + a = 1$. If $a \cdot b' \neq 0$, write $a \cdot b' = S(t)$; then $S(t) + a = S(t + a) = 1 = S(0)$, so $t + a = 0$, hence $a = 0$ by [Lemma 3.9](#), a contradiction. Thus $a \cdot b' = 0$, so $b' = 0$ by [Lemma 3.10](#), and $b = 1$. Then $a \cdot 1 = a = 1$ by [Lemma 3.7](#). $\square$

Lemma 3.13 (Cancellation in a self-product). *If $n \neq 0$ and $n \cdot k = n$, then $k = 1$.*

Proof. Write $k = S(k')$ (since $k = 0$ would give $n \cdot 0 = 0 \neq n$). Then $n \cdot k = (n \cdot k') + n = n = n + 0$, so $n \cdot k' = 0$ by [Lemma 3.8](#). Hence $k' = 0$ by [Lemma 3.10](#), and $k = 1$. $\square$

Remark 3.14. Full left/right distributivity and the associativity and commutativity of multiplication follow the same inductive pattern as [Lemmas 3.4](#) and [3.5](#); see [1]*I. The lemmas above are the only multiplicative identities invoked below, so we stop here.

4. ORDER

Definition 4.1. Write $m \leq n$ if and only if $\exists k$ with $m + k = n$, and $m < n$ if and only if $m \leq n$ and $m \neq n$.

Lemma 4.2 (Order toolkit). *For all a, b, c :*

- (i) $a \leq a$; $\leq$ is transitive; $0 \leq a$;
- (ii) $a \leq b$ if and only if $a + c \leq b + c$;
- (iii) if $a \leq b$ and $b \leq a$, then $a = b$;
- (iv) $a \neq S(a)$, and hence $a < S(a)$;
- (v) $a < b$ if and only if $\exists k \neq 0$ with $a + k = b$;
- (vi) if $a \leq b$, then $S(a) \leq S(b)$;
- (vii) if $a < b$ and $b < c$, then $a < c$;
- (viii) if $a < b$, then $S(a) \leq b$.

Proof. (i) Witness 0; add witnesses for transitivity; witness a for $0 \leq a$. (ii) Rearrange using Lemmas 3.4 and 3.5; the converse uses Lemma 3.8. (iii) If $a + k = b$ and $b + \ell = a$, then $a + (k + \ell) = a$, so $k + \ell = 0$ by Lemma 3.8, hence $k = \ell = 0$ by Lemma 3.9. (iv) Let $A = \{n : n \neq S(n)\}$. Then $0 \in A$ by Axiom 2.2. If $n \in A$ and $S(n) = S(S(n))$, then $n = S(n)$ by Axiom 2.3, a contradiction. So $A = \mathbb{N}$. With $a + 1 = S(a)$ we get $a < S(a)$. (v) Immediate from Lemma 3.8. (vi) If $a + k = b$, then $S(a) + k = S(b)$ by Lemma 3.6. (vii) Combine nonzero witnesses; their sum is nonzero by Lemma 3.9. (viii) Write $b = a + S(k') = S(a) + k'$ by Lemma 3.6. $\square$

Lemma 4.3 (Trichotomy). *For all m, n , exactly one of $m < n$, $m = n$, $m > n$ holds.*

Proof. Existence. Fix m and induct on n . For $n = 0$: if $m = 0$ we are done; if $m \neq 0$ then $0 < m$ by Lemma 4.2(i),(v). Assuming the claim for n : if $m < n$ then $m < S(n)$ by Lemma 4.2(vii),(iv); if $m = n$ then $m < S(n)$ by Lemma 4.2(iv); if $n < m$ then $S(n) \leq m$ by Lemma 4.2(viii), so $m = S(n)$ or $m > S(n)$.

Exclusivity. Pairs involving equality contradict $m \neq n$. If $m < n$ and $n < m$, write $m + k = n$ and $n + \ell = m$ with $k, \ell \neq 0$; then $m + (k + \ell) = m$, so $k + \ell = 0$, contradicting Lemma 3.9. $\square$

Lemma 4.4. *One has $n < 2$ if and only if $n = 0$ or $n = 1$. Consequently, $n > 1$ if and only if $n \geq 2$.*

Proof. If $n = 0$ or $n = 1$, then $n < 2$ by Lemma 4.2(iv),(vii). Conversely, if $n < 2$ and $n \neq 0$, then $n = S(m)$ and $m + 1 < 1 + 1$, so $m < 1$ by Lemma 4.2(ii). The only number < 1 is 0, so $n = 1$. For the second claim: if $n \geq 2$ then $n > 1$; if $n > 1$ then $n \neq 0, 1$, so $\neg(n < 2)$, hence $n \geq 2$ by trichotomy. $\square$

5. DIVISIBILITY

Definition 5.1. Write $d \mid n$ if $\exists k$ with $n = d \cdot k$.

Lemma 5.2 (Divisibility toolkit). (i) $a \mid a$, $1 \mid a$, and $a \mid 0$;

- (ii) if $a \mid b$ and $b \mid c$, then $a \mid c$;
- (iii) if $d \mid n$ and $n \neq 0$, then $d \neq 0$ and $d \leq n$;
- (iv) if $n = d \cdot k$ with $d > 1$ and $k > 1$, then $1 < d < n$ and $1 < k < n$.

Proof. (i) Use Lemma 3.7 and (3.3). (ii) Compose the witnesses. (iii) If $d = 0$ then $n = 0$ by (3.3). Since $n \neq 0$ we have $k \neq 0$ by Lemma 3.10, so $k = S(k')$ and $n = (d \cdot k') + d$, whence $d \leq n$. (iv) Neither factor is 0. The inequalities $d > 1$ and $k > 1$ are hypotheses. If $d = n$, then $n \cdot k = n$, so $k = 1$ by Lemma 3.13, a contradiction. Thus $d \neq n$. Combined with (iii) and trichotomy, $d < n$. For the cofactor: write $d = S(d')$ with $d' \neq 0$ (since $d > 1$). Then

$$n = d \cdot k = S(d') \cdot k = d' \cdot k + k$$

by Lemma 3.11. Here $d' \cdot k \neq 0$ by Lemma 3.10, so $k < n$ by Lemma 4.2(v). The same argument with the roles of the displayed summands reversed is unnecessary: we already have $1 < k$ by hypothesis. $\square$

6. PRIMES, COMPOSITES, AND EVEN NUMBERS

Definition 6.1. A number p is *prime* if $p > 1$ and every divisor of p is 1 or p .

Definition 6.2. A number n is *composite* if $n > 1$ and n is not prime.

Definition 6.3. A number n is *even* if $2 \mid n$.

Remark 6.4. Definition 6.1 is the irreducible condition in the multiplicative monoid of positive natural numbers: by Lemma 3.12 the only unit is 1, so a prime is precisely a nonunit with no nontrivial factorization. We never use Euclid's lemma $p \mid ab \Rightarrow p \mid a \vee p \mid b$.

Lemma 6.5. A number $n > 1$ is composite if and only if there exist d, k with $n = d \cdot k$, $1 < d < n$, and $1 < k < n$.

Proof. If such d, k exist, then $d \mid n$ with $d \neq 1, n$, so n is not prime. Conversely, if n is composite then some divisor d satisfies $d \neq 1$ and $d \neq n$. By Lemma 5.2(iii) and Lemma 4.4, one has $1 < d < n$. Write $n = d \cdot k$. Then $k \neq 0$ (since $n \neq 0$) and $k \neq 1$ (else $d = n$). Thus $k > 1$, and Lemma 5.2(iv) yields $1 < k < n$. $\square$

Example 6.6. (a) 0 is even ($0 = 2 \cdot 0$) but not prime, and not composite.

(b) 1 is neither even, prime, nor composite.

(c) 2 is an even prime (Proposition 6.8 below).

(d) 3 is prime: any divisor d satisfies $d \leq 3$ and $d \neq 0$, so $d \in \{1, 2, 3\}$, and $2 \nmid 3$ because $3 \neq 2 \cdot 0$ and $3 \neq 2 \cdot 1$.

(e) 4 is composite: $4 = 2 \cdot 2$ with $1 < 2 < 4$.

Lemma 6.7. If $d \mid 2$, then $d = 1$ or $d = 2$.

Proof. By Lemma 5.2(iii), $d \neq 0$ and $d \leq 2$. By Lemma 4.4, $d = 1$ or $d = 2$. $\square$

Proposition 6.8. The number 2 is prime and even.

Proof. Evenness: $2 = 2 \cdot 1$. Primality: $2 > 1$, and every divisor is 1 or 2 by Lemma 6.7. $\square$

Lemma 6.9 (Halving). If $n > 2$ and $n = 2 \cdot k$, then $1 < k < n$.

Proof. If $k = 0$ then $n = 0$; if $k = 1$ then $n = 2$; both contradict $n > 2$. Thus $k > 1$. Also $2 > 1$, so Lemma 5.2(iv) gives $k < n$. $\square$

Lemma 6.10. If $n < 2$, then n is not prime.

Proof. By Lemma 4.4, $n = 0$ or $n = 1$; neither satisfies $n > 1$. $\square$

7. THE UNIQUENESS ARGUMENT

Theorem 7.1 (Even numbers above two are composite). If $n > 2$ and n is even, then n is composite.

Proof I (definitional). Since n is even, $2 \mid n$. Also $2 \neq 1$ by Lemma 2.6, and $2 \neq n$ because $n > 2$. Thus n has a divisor other than 1 and n . Since $n > 1$, n is composite. $\square$

Proof II (factorization). Write $n = 2 \cdot k$. By Lemma 6.9, $1 < k < n$. By Lemma 6.5, n is composite. $\square$

Remark 7.2. Proof I applies the divisor clause to the factor 2; Proof II exhibits the cofactor k as a proper factor. These are different arithmetic ideas. Packaging Proof I as a contradiction, or routing it through “if primes p, q satisfy $p \mid q$ then $p = q$,” does not yield an independent proof.

Corollary 7.3. *If p is prime and even, then $p = 2$.*

Proof. By [Definition 6.1](#), $p > 1$, so $p \geq 2$ by [Lemma 4.4](#). If $p > 2$, then [Theorem 7.1](#) says p is composite, contradicting primality. By trichotomy, $p = 2$. $\square$

Lemma 7.4. *The following are equivalent, and neither asserts that 2 is prime:*

- (i) *every even $n > 2$ is not prime;*
- (ii) *every even prime equals 2.*

Proof. (i) $\Rightarrow$ (ii): if p is an even prime and $p > 2$, contradict (i); since $p > 1$, get $p = 2$. (ii) $\Rightarrow$ (i): if $n > 2$ is even and prime, then (ii) forces $n = 2$. $\square$

Theorem 7.5 (Uniqueness of the even prime). *The number 2 is an even prime, and every even prime equals 2. Equivalently: there is no even prime $n < 2$, there is no even prime $n > 2$, and 2 is an even prime.*

Proof. [Proposition 6.8](#) supplies the witness. [Lemma 6.10](#) rules out primes < 2 . [Theorem 7.1](#) rules out even primes > 2 . [Corollary 7.3](#) gives uniqueness-only. The equivalence of the two formulations is trichotomy on any even prime. Note that uniqueness-only alone does not yield the witness; one needs [Proposition 6.8](#) (or an equivalent) to obtain existence. $\square$

Corollary 7.6. *There exists a unique even prime, namely 2.*

8. COMPANION LEAN FORMALIZATION

A companion development in LEAN 4 (stdlib only) lives in `lean/` and is built by `lake build && lake test`. It checks the uniqueness layer on kernel `Nat`:

Paper	Lean
Definitions 6.1 to 6.3	<code>IsPrime</code> , <code>IsComposite</code> , <code>IsEven</code>
Proposition 6.8	<code>isPrime_two</code> , <code>isEven_two</code>
Proof I / Corollary 7.3	<code>eq_two_of_isPrime_of_isEven</code>
Proof II / Theorem 7.1	<code>isComposite_of_gt_two_of_isEven_factorization</code>
Lemma 6.10	<code>not_isPrime_of_lt_two</code>
Theorem 7.5 (existence+uniqueness)	<code>UniqueEvenPrime</code> , <code>existsUnique_even_prime</code>
uniqueness-only	<code>TargetClaim</code> / <code>EveryEvenPrimeEqTwo</code>

Lean does *not* reconstruct $\mathbb{N}$ from the Peano axioms. It assumes kernel `Nat` and uses automation for routine inequalities. The Peano narrative of [Sections 2 to 4](#) is therefore checked by the reader; the uniqueness argument relative to Lean's $\mathbb{N}$ is checked by the kernel. See `lean/README.md`.

REFERENCES

- [1] E. Landau, *Foundations of Analysis*, Chelsea, New York, 1951.